

Dr. Thejus Mahajan
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Hamburg, 10 March 2026

Ms. Anna Sacher
Institute of Computational Biology
Helmholtz Munich
Ingolstädter Landstraße 1
85764 Neuherberg

Staff Scientist (f/m/x) in Computational Biology (Job ID: 102868)

Dear Ms. Sacher,

Bridging rigorous physical science computing with modern biomedical workflows has defined my research career. The opportunity to shape scientific strategy and elevate methodological standards at the Computational Health Center aligns perfectly with my drive to engineer reproducible single-cell pipelines. My transition from a Ph.D. in computational systems modeling to bioinformatics pipeline and ML engineering makes me a highly suitable candidate to drive your single-cell and computational biology projects.

My career has been defined by bridging rigorous physical science computing with modern biomedical workflows, and I have a proven track record of mentoring and coordinating interdisciplinary teams. Throughout my post-doc and recent roles, I have spearheaded complex research designs while guiding junior colleagues and students in advanced computational techniques. At my current internship with HealthTwiSt GmbH, I took ownership of code quality for Germany's interventional radiology quality registry. I externalized hardcoded logic, verified byte-level reproducibility, and reduced a 1,834-line production pipeline by 26%—establishing the exact kind of reproducible workflows and shared best practices you require.

Your need for expertise in modern scientific computing frameworks (JAX, PyTorch, scikit-learn) aligns perfectly with my core technical strengths:

- **Scientific Computing & AI (JAX/PyTorch/scikit-learn):** During my post-doc, I led the translation of a legacy Fortran model into a high-performance Python engine using **Google JAX**. By engineering branchless logic and vectorized operations, I achieved direct compatibility with **Google TPUs and GPUs** for massive parallelization. Because JAX shares core paradigms with **PyTorch**, I possess the deep, under-the-hood expertise required to scale AI architectures. I am also actively expanding my applied ML capabilities with scikit-learn and Tidymodels architectures.
- **Mentorship & Scientific Strategy:** I have extensive experience coordinating research initiatives and preparing scientific content. From coaching state-level teams as an educator to spearheading evolutionary systems modeling at Helmholtz Hereon, I excel at translating complex quantitative methods to diverse, interdisciplinary teams.
- **Reproducible Workflows & Single-Cell Concepts:** Highly proficient in building reproducible, containerized workflows (Nextflow/DSL2) and processing multi-omic data. I bring a strong understanding of the statistical modeling required to extract signal from high-dimensional biological datasets like transcriptomics.

I am deeply motivated by Helmholtz Munich's mission to bridge fundamental methodological development with translational biomedicine. My robust mathematical background and dedication to reproducible engineering will allow me to elevate your lab's computational standards while mentoring the next generation of researchers. I welcome the opportunity to discuss further and am available for an interview.

Sincerely



Dr. Thejus Mahajan

THEJUS MAHAJAN PH.D.

Bioinformatics & Data Scientist

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Hamburg, Germany

*30.09.1992 (Talikulam, India)

Married



PROFILE

- **Focus:** Large-scale data analysis, bioinformatics pipeline development, and statistical modeling.
- **Core Stack:** R (Bioconductor, Tidymodels), Python (Pandas, JAX, Scikit-learn), Nextflow (DSL2), BASH.
- **Current Objective:** Applying reproducible computational workflows and AI methodologies to advance translational collaborations in biomedicine.

EXPERIENCE

Internship – Bioinformatics & Data Pipeline Engineering

HealthTwist GmbH

02/2026 - 04/2026

Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn

- Refactored a 1,834-line medical data pipeline (R/tidymodels) for Germany's interventional radiology quality registry (DeGIR); 143K rows × 1,920 columns.
- Externalized 257 hardcoded rules into 8 config files; 26.5% code reduction with byte-level output verification (`identical()`).
- Identified 2 latent bugs through systematic code analysis; documented without altering pipeline behavior.
- Machine Learning workflows using the Tidymodels framework in R.

Further Training – Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH

08/2025 - 02/2026

Berlin, Germany

Application-related Bioinformatics and Biostatistics

- Programming methodology: Shell (BASH), Python (object-oriented), R.
- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Nextflow, Galaxy); structural bioinformatics.
- Clinical biostatistics (R/Bioconductor, ANOVA, PCA, HCA, survival analysis); GDPR/BDSG compliance.

Guest Scientist

Helmholtz-Zentrum Hereon, Ökosystemmodellierung

05/2025 - 10/2025

Geesthacht, Germany

Computational Biologist & Systems Modeler

- Spearheaded research modeling evolutionary processes in marine ecosystems.
- Applied advanced computational techniques to simulate long-term environmental impacts.

Job Search & German Language Learning

Hamburg, Germany

02/2025 - 04/2025

Hamburg, Germany

Active job search and German language acquisition (Goethe B1 certification).

Post-doctoral Scientist

University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences

08/2021 - 01/2025

Hamburg, Germany

Marine Ecosystem Modelling

- *Elternzeit (Parental Leave) from 01/2024 to 12/2024.*

STRENGTHS



Data Pipeline Engineering

Refactored a production medical data pipeline (1,834 lines, 143K rows), achieving 26.5% code reduction with zero behavioral regression. Byte-level verification after every change.



Complex Problem-Solving

Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector, recalculating results using multi-sensor data.



Bioinformatics & Biostatistics

Bridged physical science expertise with modern biological workflows. Skilled in complex multidimensional array operations, JAX/GPU acceleration, sequence alignment, and multivariate statistical modeling using R (Bioconductor) and Python.

TECHNICAL SKILLS

Programming

Python (JAX, Pandas, NumPy, Scikit-learn, Biopython), R (Bioconductor, Tidymodels), SQL, BASH

Bioinformatics

Nextflow (DSL2), NGS Analysis, Multiple Sequence Alignment (MAFFT), Sequence Trimming (trimAl), NCBI / Entrez

Biostatistics & ML

Deep Learning Foundations (TensorFlow, PyTorch concepts via JAX), Multivariate Analysis, PCA, Clustering, Scikit-learn, Tidymodels

Data Engineering

Hardware Acceleration (TPU/GPU), Large dataset processing (HDF5, NetCDF), Pipeline development

Tools

Linux/Unix, Git, HPC clusters, Conda

TRAINING / COURSES

IBM: Machine Learning with Python; Introduction to AI

Coursera (in progress)

High Performance Computing (HPC) Training

Jülich Supercomputing Center (JSC), Forschungszentrum Jülich

- **Model Development:** Developed the novel “Cyanobacteria Life Cycle” (CLC) model within the ERGOM framework for climate warming projections.
- **High-Performance Computing & ML:** Led the translation of a legacy Fortran continuous model into a high-performance Python simulation engine. Utilized **Google JAX** vectorization and branchless logic for massive parallelization (direct compatibility with **TPUs and GPUs**).
- This architecture established a strong foundation in tensor operations underpinning deep learning frameworks like **PyTorch** and **TensorFlow**.

Job Search

Kerala, India

04/2021 - 07/2021

Kerala, India

Relocation from India to Germany and active job search.

Physics Educator

Tutorwaves Solutions Inc / Freelance

10/2018 - 03/2021

Kerala, India

Prepared physics content for competitive examinations and coached state-level chess teams.

Ph.D. Researcher in Astrochemistry

University of Paris-Saclay

10/2015 - 09/2018

Orsay, France

Led atomic collision experiments, analyzed data, and modeled outcomes.

- Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
- C++ signal processing, algorithm development, and molecular fragmentation modeling (KIDA database).

EDUCATION

Ph.D. in Astrochemistry

Université Paris-Saclay, Institute of Molecular Sciences (ISMO) Orsay, France

09/2015 - 09/2018

Orsay, France

M.Sc. in Physics

National Institute of Technology Calicut, India

07/2012 - 01/2015

Calicut, India

B.Sc. in Physics

University of Calicut, India

06/2009 - 04/2012

Thrissur, India

REFERENCES

Dr. Andreas Busjahn

abusjahn@healthtwist.de

Prof. Dr. Kai Wirtz

kai.wirtz@hereon.de

Prof. Dr. Elisa Schaum

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Dr. Thejus Mahajan

MITx: 7.00x Introduction to Biology - The Secret of Life

Comprehensive coursework on genetics, biochemistry, and molecular biology.

OTHER SELECTED PROJECTS

Hepatitis Delta Virus Pipeline • GitHub

Nextflow (DSL2) pipeline for viral sequence analysis: NCBI Entrez download, MAFFT alignment, trimAl trimming, Conda-managed dependencies.

JAX/TPU High-Performance Simulator • GitHub

Ported a legacy Fortran model to a modular Python architecture using Google JAX. Engineered branchless, vectorized logic for massive TPU/GPU parallelization.

Introduction to Tidymodels • GitHub

Teaching script for ML in R: recipes, parsnip, workflows, and yardstick on the Palmer Penguins dataset.

Berlin Tree Analysis • GitHub

Interactive Jupyter notebook analyzing 5 years of tree planting data in Berlin using Python, Pandas, and Seaborn.

PUBLICATIONS

5 peer-reviewed articles • Google Scholar Profile

LANGUAGES

English



Proficient

German



B1 - Goethe Certified

French



Intermediate

Malayalam



Native

Tamil



Proficient

Hindi



Intermediate